

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester III Examination, 2023-24

Paper No. MTB-302: Differential Equations

Time: 3 Hours

Full Marks: 70

Ref. No.: 067789

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Define the order and degree of a differential equation.
- (b) Obtain the relation in a, b, c (constants) when $y = xe^{ax+bx^2+cx^3}$ is a solution of $y'' + P(x)y' + Q(x)y = 0$.
- (c) Find an integrating factor for $y(axy + e^x)dx - e^x dy = 0$.
- (d) Solve $\frac{dy}{dx} + ay = e^{mx}$ where a, m are constants.
- (e) Obtain the differential equation of the family of plane curves $cy = x^2 + c^2$.
- (f) Let $y(x)$ be a solution of $\frac{dy}{dx} + 3y = 4$, $y(0) = 3$. Find $\lim_{x \rightarrow \infty} y(x)$.
- (g) Find the value of $\frac{1}{(D+a)^2} \sin(ax)$ where $D = \frac{d}{dx}$.
- (h) Find the condition for $y = e^{ax}$ to be a solution of $\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0$.
- (i) Write a 2nd order ODE with $x = 1$ as a regular singular point and $x = 0$ as an irregular singular point.
- (j) Express $x^3 + 3x^2$ in terms of Legendre polynomials.
- (k) Prove that $J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x$.

Question 2.

- (a) Solve the IVP: $\frac{dy}{dx} + y \cot x = \cos^2 x (1 + y^2 \sin^2 x)$, $y\left(\frac{\pi}{2}\right) = 1$. [6]
- (b) Solve: $xyp^2 + p(3x^2 - 2y^2) - 6xy = 0$ where $p = \frac{dy}{dx}$. [6]

Question 3.

(a) Solve: $\frac{d^4 y}{dx^4} - \frac{d^2 y}{dx^2} = e^{-x} + 2 + x.$ [6]

(b) Solve: $\frac{d^2 y}{dx^2} + a^2 y = \sec(ax).$ [6]

Question 4.

(a) If $f(D) = \sum_{r=0}^n a_r D^r$, establish:

$$\frac{1}{f(D)}[e^{ax} f_2(x)] = e^{ax} \frac{1}{f_1(-a^2) + a f_2(-a^2)} \left[\frac{x^n}{n!} \right]$$

where $f(D) = f_1(D^2) + D f_2(D^2).$ [6]

(b) Using variation of parameters, solve: $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x^2 \log x, x > 0.$ [6]

Question 5. Solve in series:

$$x \frac{d^2 y}{dx^2} + \frac{dy}{dx} + xy = 0$$

[14]

Question 6.

(a) Prove that $\int_{-1}^1 P_m(x) P_n(x) dx = 0$ if $m \neq n$, where $P_n(x)$ are Legendre polynomials. [7]

(b) Prove that $\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$ where $J_n(x)$ is the Bessel function. [5]

Question 7.

(a) Show that $P_n(x)$ is the coefficient of z^n in the expansion of $(1 - 2xz + z^2)^{-1/2}$ in ascending powers of z , $|x| < 1, |z| < 1.$ [7]

(b) Without using any recurrence relation, prove that $J_{3/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right).$ [5]